



SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED



LOCATION: Jetty-1	AREA: B
PERMIT TO WORK No.: -	TYPE OF SCAFFOLD: Independent
SCAFFOLD DRAWING No: PMS-WA-ARCO-025	
RISK CLASSIFICATION: ☒ High ☐ Medium ☐ Low	

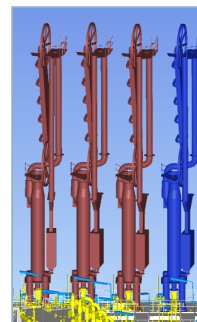
SCAFFOLD TO BE INSPECTED BY PIC BEFORE TAGGING

BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of an Independent scaffold required for maintenance of the loading arm, equipment tag Z-3408 at Jetty-1. The scaffold envelope is 3m x 1.5m x 15m (length x width x height) and is configured with standards, ledgers, transoms, bracing, platforms, and support/tie arrangements as indicated on the approved drawing.

SAFE WORKING LOAD (SWL): 1.5 kN/m²

SITE / EQUIPMENT PHOTO



Total Horizontal Load in the X Direction (F_x): 1.352 kN

Total Horizontal Load in the Z Direction (F_z): 16.298 kN

Role	ID	Name	Signature	Date
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	Project Name:	SCAFFOLD DESIGN REPORT		
	Scaffold Type:	Independent	Calculation Sheet	
	Document Number:	PMS-WA-ARCO-025	Prepared By: Nnamani, Ifeanyi	

TABLE OF CONTENTS

CALCULATION SHEETS

General Note, Material Properties & Modelling Philosophy

Load Cases

Wind Load Calculation (EN 1991-1-4:2005)

Load Combinations

Static Check - Utilization Ratio Summary

Scaffold Tubes Connection Stability Check

Deflection Check Results

Conclusion, Recommendations & Installation Notes

References & Standards

ENGINEERING DRAWINGS

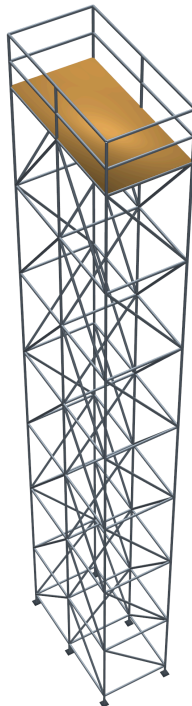
Engineering Drawing - Front View

Engineering Drawing - Side View

Engineering Drawing - Plan View

Engineering Drawing - 3D View

3D ISOMETRIC VIEW



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GENERAL NOTE

BS EN 12811-1, 4.2.1.2 / 10.3.2.2

Scaffold tube material has a minimum nominal yield strength of 235 N/mm² (BS EN 12811-1, 4.2.1.2). A partial safety factor $\gamma_M = 1.1$ (BS EN 12811-1, 10.3.2.2) has been applied, giving a design resistance of 213.6 N/mm² used in member capacity checks. This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 - Design of Steel Structures).

Material Properties

48.3 mm outside diameter x 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6**.

Modelling Philosophy

BS EN 12811-1, 10.2.3.2 / 10.2.3.3

A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint ($K_{FX}/K_{FZ} = 17.338$ kN). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot.

LOAD CASES

Dead Load / Self-weight

Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m³**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

Live Load on Platforms

BS EN 12811-1

Platform live loading is applied to the scaffold working platforms as member UDLs in the STAAD model. The load intensity is multiplied by the tributary width of each supporting member to obtain the applied member line load. (LC 2).

Member	Tributary Width (m)	Load Intensity (kN/m ²)	Member Load (kN/m)
Members 100, 103 (Edge)	$1.500 / 2 = 0.750$	1.500	1.125
Member 104 (Interior)	$(0.750 + 0.750) = 1.500$	1.500	2.250

Load Intensity = **1.50 kN/m²** — BS EN 12811-1 Class 2

Member Load = Load Intensity × Tributary Width. Edge members carry half the adjacent bay width; interior members carry half the bay from each side.

Governing load intensity (SWL) = **1.5 kN/m²** — BS EN 12811-1 Class 2.

Total applied platform live load = **21.0 kN**.

Handrail Load

Handrail loads were applied where required by the scaffold configuration and modelled independently in the global X and Z directions.

Handrail Load (HLX) = **3.6 kN** (member UDL 0.15 kN/m)

Handrail Load (HLZ) = **4.8 kN** (member UDL 0.15 kN/m)

Wind Load

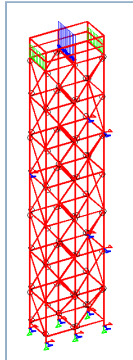
EN 1991-1-4:2005

Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.073468 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

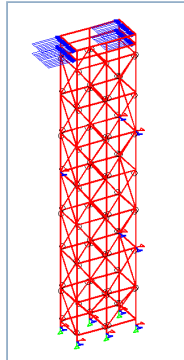
Total Wind Load - X Axis (WLX) = **54.05 kN**

Total Wind Load - Z Axis (WLZ) = **58.06 kN**

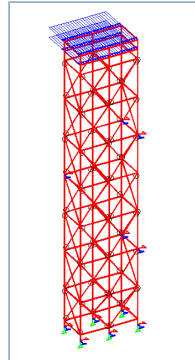
Load Diagrams



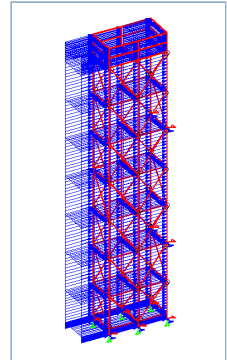
Live Load (LL)



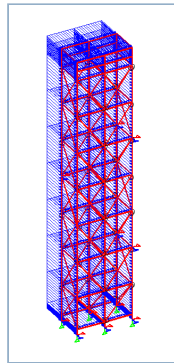
Handrail Load X (HLX)



Handrail Load Z (HLZ)



Wind Load X (WLX)



Wind Load Z (WLZ)

WIND LOAD CALCULATION - EN 1991-1-4:2005

Reference	Calculation	Value	Unit
	Basic wind speed (3-second gust)		
	V3 =	36.0	m/s
	<i>Eurocode $V_{b,0}$ is the characteristic 10-minute mean wind velocity</i>		
Durst, C.S. (1960)	Convert 3-second gust to 10-minute mean wind velocity		
	V3 / V3600 =	1.52	
	V600 / V3600 =	1.06	
	V600 = (V600/V3600) x (V3600/V3) x V3 =	25.11	m/s
	Basic wind speed $V_b = V600 =$	25.11	m/s
EN 1991-1-4:2005	Roughness Length (z0) - Terrain Category 0	0.003	m
	Terrain category II (z0,ii)	0.05	
EN 1991-1-4, Table 4.1	Z _{min}	1.0	m
	z - height of structure under consideration	15.0	m
EN 1991-1-4, Sec 4.3	C _{r0} (z) - Roughness factor	1.0	

Reference	Calculation	Value	Unit
EN 1991-1-4, Sec 4.3	$C_o(z)$ - Orography factor	1.0	
EN 1991-1-4, Sec 4.3	Terrain factor $K_r = 0.19 \times [z_0/z_{0,ii}]^{0.07}$	0.156	
EN 1991-1-4, Table 4.1	$z_{min} < z \rightarrow$ Case 1 applies		
Wind Turbulence			
	$C_r(z) = K_r \times \ln(z/z_0)$	1.329	
	Mean velocity $V_m(z) = C_r(z) \times C_o(z) \times V_b$	33.365	m/s
EN 1991-1-4, Sec 4.4	K_1	1.0	
	Turbulence intensity $I_v(z) = k_1 / [C_o(z) \times \ln(z/z_0)]$	0.1174	
Peak Velocity Pressure			
	Density of air ρ	1.25	kg/m ³
EN 1991-1-4, Sec 4.5	$q_p(z) = [1 + 7 \times I_v(z)] \times 0.5 \times \rho \times V_m(z)^2$	1267.5573	N/m ²
Wind Load = $C_f \times q_p \times A_f$			
$C_f =$		1.2	
$q_p =$		1.267557	kN/m ²
$A_f =$	Tube outer diameter (48.3 mm)	0.0483	m
Wind Load UDL =		0.073468	kN/m

LOAD COMBINATIONS

BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use gamma = **1.5**;
SLS (characteristic) combinations use gamma = **1.0**.

DL = Dead Load | **LL** = Platform Live Load | **HLX/HLZ** = Handrail X/Z | **WLX/WLZ** = Wind X/Z

LC No.	Combination	Component Loadings & Factors
7	SLS DL + LL	LC1 x 1.0 + LC2 x 1.0
8	SLS DL + LL + HLX	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0
9	SLS DL + LL + HLZ	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0
10	SLS DL + WLX	LC1 x 1.0 + LC5 x 1.0
11	SLS DL + WLZ	LC1 x 1.0 + LC6 x 1.0
12	SLS DL + LL + WLX	LC1 x 1.0 + LC2 x 1.0 + LC5 x 1.0
13	SLS DL + LL + WLZ	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0
14	SLS DL + LL + WLZ + HLX	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0 + LC3 x 1.0
15	SLS DL + LL + WLZ + HLZ	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0 + LC4 x 1.0
16	ULS 1.5DL + 1.5LL	LC1 x 1.5 + LC2 x 1.5
17	ULS 1.5DL + 1.5LL + 1.5HLX	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5
18	ULS 1.5DL + 1.5LL + 1.5HLZ	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5
19	ULS 1.5DL + 1.5LL + 1.5WLX	LC1 x 1.5 + LC2 x 1.5 + LC5 x 1.5
20	ULS 1.5DL + 1.5LL + 1.5WLZ	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5
21	ULS 1.5DL + 1.5LL + 1.5WLZ + 1.5HLX	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5 + LC3 x 1.5
22	ULS 1.5DL + 1.5LL + 1.5WLZ + 1.5HLZ	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5 + LC4 x 1.5
23	ULS 1.5DL + 1.5WLX	LC1 x 1.5 + LC5 x 1.5
24	ULS 1.5DL + 1.5WLZ	LC1 x 1.5 + LC6 x 1.5
25	ULS DL + 1.5WLX	LC1 x 1.0 + LC5 x 1.5
26	ULS DL + 1.5WLZ	LC1 x 1.0 + LC6 x 1.5

STATIC CHECK PARAMETERS - UTILIZATION RATIO SUMMARY

Maximum Utilization Ratio = 0.352 at Member No. **1500**, Load Combination **22**, Clause **EC-6.2.9.1**. **PASS** - All members adequate; UC ratio below unity.

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
1500	PIP483.6	PASS	EC-6.2.9.1	0.352	22	0.26	C
1310	PIP483.6	PASS	EC-6.3.3-661	0.312	22	11.97	C
1482	PIP483.6	PASS	EC-6.2.9.1	0.312	22	0.19	C
477	PIP483.6	PASS	EC-6.3.3-662	0.310	22	13.40	C
690	PIP483.6	PASS	EC-6.2.9.1	0.304	22	0.91	T
1304	PIP483.6	PASS	EC-6.3.3-661	0.304	22	11.66	C
1497	PIP483.6	PASS	EC-6.2.9.1	0.301	22	0.20	C
462	PIP483.6	PASS	EC-6.3.3-662	0.300	22	12.91	C
1487	PIP483.6	PASS	EC-6.2.9.1	0.284	22	0.27	C
688	PIP483.6	PASS	EC-6.3.3-662	0.265	22	0.28	C
1309	PIP483.6	PASS	EC-6.3.3-661	0.265	22	10.08	C
672	PIP483.6	PASS	EC-6.3.3-662	0.255	22	1.73	C
678	PIP483.6	PASS	EC-6.3.3-662	0.251	22	1.80	C
666	PIP483.6	PASS	EC-6.2.9.1	0.247	22	0.03	C
675	PIP483.6	PASS	EC-6.2.9.1	0.247	22	0.21	T
1490	PIP483.6	PASS	EC-6.2.9.1	0.245	22	0.21	C
478	PIP483.6	PASS	EC-6.3.3-662	0.244	22	10.04	C
463	PIP483.6	PASS	EC-6.3.3-662	0.240	22	9.82	C
647	PIP483.6	PASS	EC-6.3.3-662	0.240	22	0.89	C
1486	PIP483.6	PASS	EC-6.2.9.1	0.240	22	0.22	C
473	PIP483.6	PASS	EC-6.3.3-662	0.223	22	8.91	C
1327	PIP483.6	PASS	EC-6.3.3-661	0.223	19	6.65	C
464	PIP483.6	PASS	EC-6.3.3-662	0.221	22	8.82	C
1303	PIP483.6	PASS	EC-6.3.3-661	0.220	22	8.32	C
1201	PIP483.6	PASS	EC-6.3.3-661	0.216	21	8.16	C
1219	PIP483.6	PASS	EC-6.3.3-661	0.215	19	6.41	C
442	PIP483.6	PASS	EC-6.2.9.1	0.207	22	0.17	T
480	PIP483.6	PASS	EC-6.3.3-662	0.206	22	8.84	C
1273	PIP483.6	PASS	EC-6.3.3-661	0.206	22	7.77	C
427	PIP483.6	PASS	EC-6.2.9.1	0.203	22	0.43	T
431	PIP483.6	PASS	EC-6.3.3-662	0.200	22	0.17	C
650	PIP483.6	PASS	EC-6.3.3-662	0.200	22	1.28	C
1213	PIP483.6	PASS	EC-6.3.3-661	0.199	19	5.86	C
1328	PIP483.6	PASS	EC-6.3.3-661	0.197	19	7.41	C
1489	PIP483.6	PASS	EC-6.2.9.1	0.197	22	0.25	T
420	PIP483.6	PASS	EC-6.3.3-662	0.196	22	8.34	C
1196	PIP483.6	PASS	EC-6.3.3-661	0.196	22	7.35	C
1485	PIP483.6	PASS	EC-6.2.9.1	0.196	22	0.24	T
452	PIP483.6	PASS	EC-6.2.9.1	0.195	22	1.94	T
646	PIP483.6	PASS	EC-6.3.3-662	0.195	22	1.10	C
1212	PIP483.6	PASS	EC-6.3.3-661	0.195	19	5.74	C
416	PIP483.6	PASS	EC-6.3.3-662	0.192	22	8.19	C
1195	PIP483.6	PASS	EC-6.3.3-661	0.190	21	7.12	C

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
1194	PIP483.6	PASS	EC-6.3.3-661	0.189	20	7.06	C
475	PIP483.6	PASS	EC-6.3.3-662	0.188	19	7.66	C
472	PIP483.6	PASS	EC-6.3.3-662	0.187	19	7.56	C
1321	PIP483.6	PASS	EC-6.3.3-661	0.186	19	5.46	C
1494	PIP483.6	PASS	EC-6.2.9.1	0.186	21	0.22	C
297	PIP483.6	PASS	EC-6.3.3-662	0.185	22	7.69	C
537	PIP483.6	PASS	EC-6.3.3-662	0.185	22	7.57	C
Showing highest 50 utilization ratios; 1021 lower-utilization members omitted.							

SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **13.400 kN** (Member 477, C). The required right-angle coupler class is selected from Table C.1 below using the slipping resistance limit.

Selected coupler class: **Class B** (capacity = 15.0 kN).

Check: 13.400 kN ≤ 15.0 kN -> **PASS**

Class B right-angle couplers are required.

Table C.1 - Characteristic Values of Resistances for Couplers (EN 12811)

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Right-angle Coupler (RA)	Slipping force F _{s,k} (kN)	10.0	15.0
	Cruciform bending moment M _{B,k} (kNm)	-	0.8
	Pull-apart force F _{p,k} (kN)	20.0	30.0
	Rotational moment M _{T,k} (kNm)	-	0.13
Swivel Coupler (SW)	Slipping force F _{s,k} (kN)	6.0	9.0
	Cruciform bending moment M _{B,k} (kNm)	-	0.8
	Pull-apart force F _{p,k} (kN)	15.0	22.5
	Rotational moment M _{T,k} (kNm)	-	0.08
Putlog / Single Coupler (PL)	Slipping force F _{s,k} (kN)	-	3.0
	Cruciform bending moment M _{B,k} (kNm)	-	0.6

Member Axial Force Summary (Top 30 Horizontal Members — ULS Code Check)

Member	Type	Status	Clause	UC Ratio	L/C	Axial Force (kN)
477	C	PASS	EC-6.3.3-662	0.310	22	13.400
462	C	PASS	EC-6.3.3-662	0.300	22	12.910
1310	C	PASS	EC-6.3.3-661	0.312	22	11.970
1304	C	PASS	EC-6.3.3-661	0.304	22	11.660
1309	C	PASS	EC-6.3.3-661	0.265	22	10.080
478	C	PASS	EC-6.3.3-662	0.244	22	10.040
463	C	PASS	EC-6.3.3-662	0.240	22	9.820
473	C	PASS	EC-6.3.3-662	0.223	22	8.910
480	C	PASS	EC-6.3.3-662	0.206	22	8.840
464	C	PASS	EC-6.3.3-662	0.221	22	8.820
420	C	PASS	EC-6.3.3-662	0.196	22	8.340
1303	C	PASS	EC-6.3.3-661	0.220	22	8.320

Member	Type	Status	Clause	UC Ratio	L/C	Axial Force (kN)
416	C	PASS	EC-6.3.3-662	0.192	22	8.190
1201	C	PASS	EC-6.3.3-661	0.216	21	8.160
1273	C	PASS	EC-6.3.3-661	0.206	22	7.770
297	C	PASS	EC-6.3.3-662	0.185	22	7.690
50	C	PASS	EC-6.3.3-661	0.073	19	7.670
475	C	PASS	EC-6.3.3-662	0.188	19	7.660
537	C	PASS	EC-6.3.3-662	0.185	22	7.570
472	C	PASS	EC-6.3.3-662	0.187	19	7.560
356	C	PASS	EC-6.3.3-662	0.179	22	7.550
522	C	PASS	EC-6.3.3-662	0.184	22	7.500
116	C	PASS	EC-6.3.3-662	0.144	22	7.490
296	C	PASS	EC-6.3.3-662	0.176	22	7.410
1328	C	PASS	EC-6.3.3-661	0.197	19	7.410
60	C	PASS	EC-6.3.3-662	0.074	22	7.400
236	C	PASS	EC-6.3.3-662	0.175	22	7.370
1196	C	PASS	EC-6.3.3-661	0.196	22	7.350
101	C	PASS	EC-6.3.3-662	0.141	22	7.300
360	C	PASS	EC-6.3.3-662	0.173	22	7.250

FRICTIONAL RESISTANCE

Max vertical reaction per support ($\text{Max } f_y$) = Resistance x 0.15 / C_F = **8.669 kN**

Coefficient of friction (C_F) = **0.3**

Frictional Resistance (FR) = $C_F \times \text{Max } f_y$ = **2.6007 kN**

Spring Stiffness = FR / 0.15 = **17.338 kN** <- applied as KFX/KFZ in STAAD model

DEFLECTION CHECK RESULTS

BS EN 12811-1, Cl 6.3 Serviceability limit state — values from STAAD.Pro SLS (unfactored) combination results.
Displacement tables are shown below.

Vertical Deflection Check

Parameter	Value	Allowable	Check
Max vertical displacement - LC 9 Member 104 Member length L = 1500.0 mm	2.21 mm	$L/100 = 1500/100 = 15.0 \text{ mm}$	PASS

Vertical Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
104	2.210	0.750	9	678	0.041	-2.527	1.693
104	2.209	0.750	7	678	0.006	-2.507	0.277
104	2.209	0.750	12	678	0.396	-2.507	0.339
104	2.208	0.750	8	679	0.131	-2.507	0.293
104	2.207	0.750	14	679	0.150	-2.516	3.342
104	2.207	0.750	15	679	0.061	-2.535	4.742
104	2.207	0.750	13	679	0.025	-2.516	3.325
100	0.980	0.625	15	1530	0.199	-1.232	1.642
103	0.977	0.875	15	1534	-0.010	-1.136	3.095
103	0.951	0.750	13	1576	-0.019	-1.201	1.682
103	0.951	0.750	14	1576	0.153	-1.161	1.741
100	0.950	0.625	9	1578	0.118	-1.225	0.710
100	0.950	0.750	8	1578	0.221	-1.277	-0.044
103	0.949	0.750	9	1580	-0.003	-1.203	1.299
103	0.949	0.750	7	1580	-0.038	-1.228	-0.115

Max = 2.21 mm (LC 9) | L = Member 104 length (1500 mm) | Allowable L/100 = 15.0 mm → **PASS**

Horizontal Deflection Check

Parameter	Value	Allowable	Check
Max horizontal displacement - LC 9 Member 93 Member length L = 2000.0 mm	1.122 mm	L/200 = 2000/200 = 10.0 mm	PASS

Horizontal Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
93	1.122	1.333	9	1782	0.091	-0.136	0.402
93	1.042	1.333	15	1919	0.157	-0.020	3.422
96	0.984	1.333	14	2033	0.025	-0.460	4.188
96	0.981	1.333	13	2037	-0.042	-0.457	4.174
96	0.905	1.333	8	2210	0.067	-0.308	1.169
96	0.904	1.333	12	2212	0.390	-0.313	1.210
96	0.903	1.333	7	2215	0.000	-0.307	1.155
93	0.860	1.333	8	2326	0.057	-0.265	-0.594
93	0.858	1.333	12	2330	0.356	-0.259	-0.554
93	0.857	1.333	7	2332	0.008	-0.268	-0.608
93	0.780	1.333	14	2565	0.123	-0.149	2.426
93	0.777	1.333	13	2573	0.074	-0.150	2.412
96	0.721	1.167	15	2774	-0.072	-0.622	5.114
96	0.641	1.333	9	3122	-0.029	-0.476	2.163
94	0.555	1.333	15	3604	0.193	-0.018	0.784

Max = 1.122 mm (LC 9) | L = Member 93 length (2000 mm) | Allowable L/200 = 10.0 mm → **PASS**

CONCLUSION

The structural analysis and design of the **Independent** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = 0.352 at Member 1500). Vertical and horizontal deflections are within their respective allowable limits of **L/100 = 15.0mm** and **L/200 = 10.0mm**. All scaffold tube coupler connections are adequate for the applied axial loads.

RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and

material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

INSTALLATION NOTES

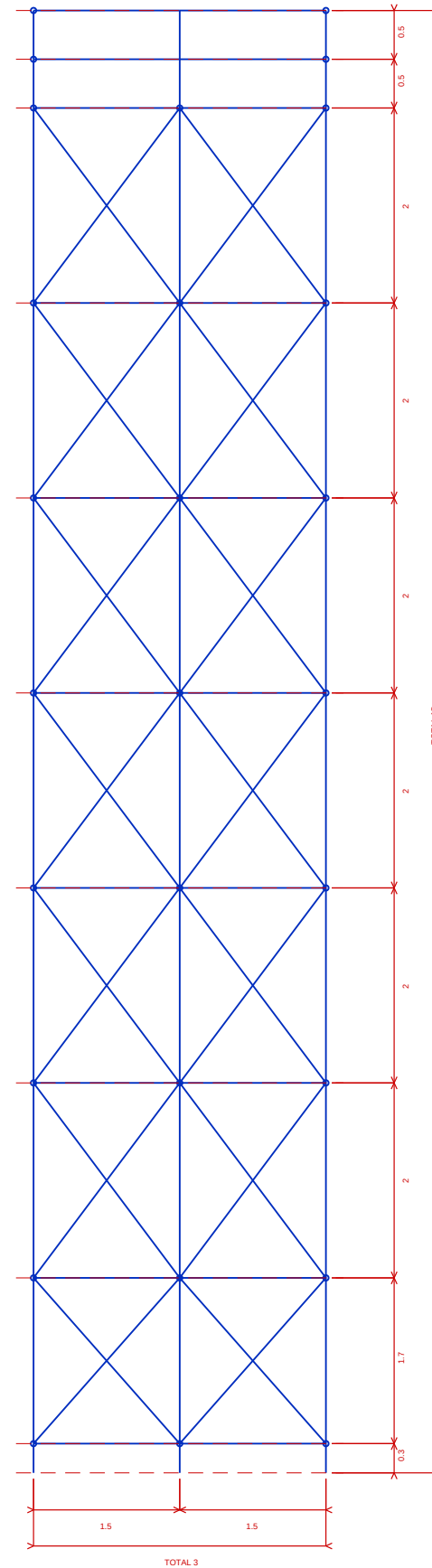
1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.
2. Confirm that the Independent scaffold configuration matches the approved drawing and is suitable for maintenance.
3. The work vicinity must be barricaded and appropriate signage placed at all access points.
4. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
5. A valid work permit must be obtained, discussed, and signed before erection commences.
6. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
7. Set out and level the scaffold base plates and sole boards in accordance with the approved working drawing.
8. Confirm the scaffold footprint/envelope is approximately 3.0m x 1.5m x 15.0m or as otherwise dimensioned on the approved drawing.
9. Erect scaffold tube standards (48.3 x 3.6 mm), ledgers, and transoms progressively from the base, checking plumb and level in both axes before proceeding to the next lift.
10. Install the lower horizontal ledgers and transoms at the kicker lift height (0.300m) to form the first stable frame.
11. Install mid-level horizontal ledgers and transoms at 2.0m height in accordance with the working drawing.
12. Install mid-level horizontal ledgers and transoms at 4.0m height in accordance with the working drawing.
13. Install mid-level horizontal ledgers and transoms at 6.0m height in accordance with the working drawing.
14. Install mid-level horizontal ledgers and transoms at 8.0m height in accordance with the working drawing.
15. Install mid-level horizontal ledgers and transoms at 10.0m height in accordance with the working drawing.
16. Install mid-level horizontal ledgers and transoms at 12.0m height in accordance with the working drawing.
17. Install mid-level horizontal ledgers and transoms at 14.0m height in accordance with the working drawing.
18. Install mid-level horizontal ledgers and transoms at 14.5m height in accordance with the working drawing.
19. Install all plan, face, and longitudinal bracing shown on the approved drawing to provide sway resistance in both principal directions.
20. Install platform boards, guardrails, mid-rails, toe boards, access arrangements, and any handrail load-resisting members required by the design.
21. Tighten and torque-check all load-bearing couplers and support connections before the scaffold is released for inspection.
22. Gravlock or box-tie the scaffold structure to the existing structure at all indicated tie points at 4.0m and 8.0m and 10.0m height as shown on the working drawing.
23. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before use.
24. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
25. All modifications made after erection must be carried out by a competent scaffold team with the approval of the scaffold engineer. End-user or third-party modification of this scaffold structure is strictly prohibited.
26. For dismantling, perform the erection steps in reverse order.

REFERENCES & STANDARDS

Reference	Title	Year
BS EN 12811-1:2003	Temporary Works Equipment - Scaffolds - Performance Requirements and General Design	2003
BS EN 1993-1-1:2005	Eurocode 3: Design of Steel Structures - General Rules and Rules for Buildings	2005
BS EN 1991-1-4:2005	Eurocode 1: Actions on Structures - Wind Actions	2005
BS EN 1990:2002	Eurocode: Basis of Structural Design — Load Combinations, Partial Factors, and Limit States (ULS/SLS/EQU)	2002
NASC TG20:13	Guide to Good Practice for Scaffolding with Tubes and Fittings	2013
Durst, C.S.	Wind Speeds Over Short Periods of Time. The Meteorological Magazine, 89, 181–186 — basis of 3-second gust to 10-minute mean conversion	1960
STAAD.Pro CONNECT	STAAD.Pro CONNECT Edition V22 - Structural Analysis and Design Software	2022

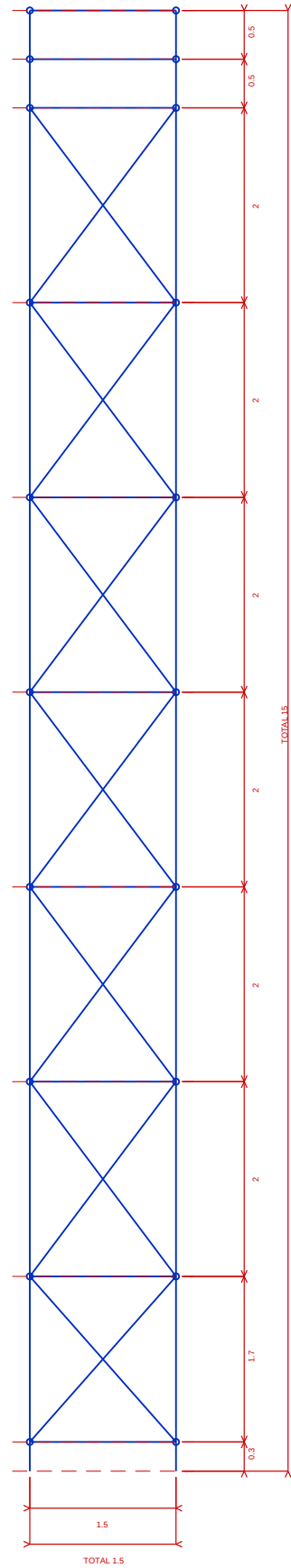
Software Used: STAAD.Pro CONNECT Edition V22 (Bentley Systems) — 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.

Report: Scaffold Structural Design Report — ARCO M&E, NLNG Bonny Island.



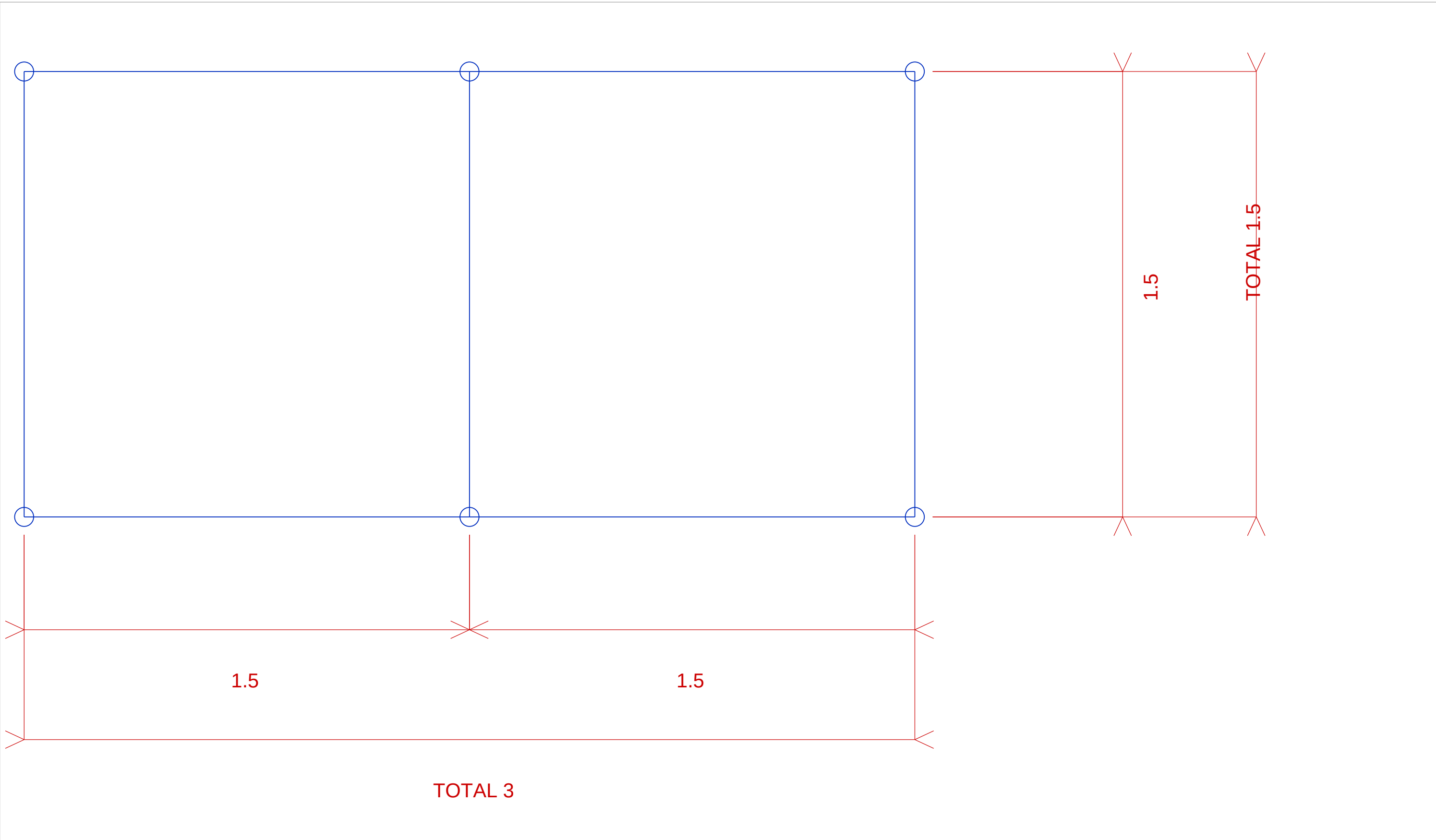
FRONT VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-025-FRONT-VIEW			 
DRAWING TITLE PMS-WA-ARCO-025 - FRONT VIEW		SCALE NTS	DATE 07-JUL-2026	REV 0	
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi	COMPANY
					SHEET 1 OF 4



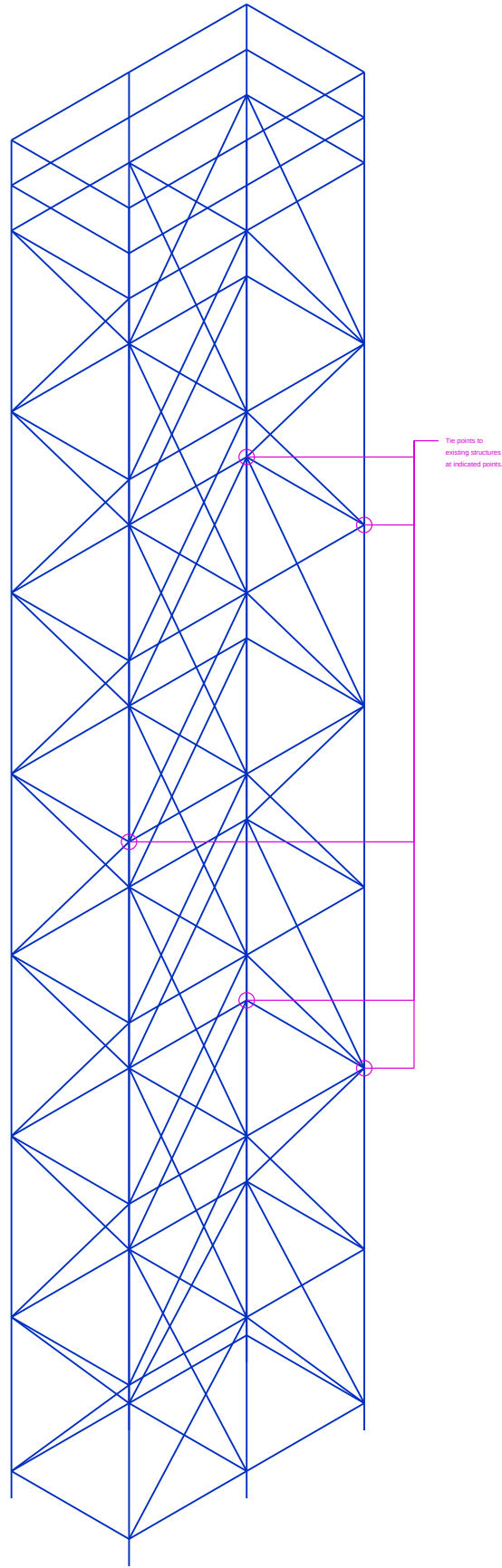
SIDE VIEW

PROJECT		DRAWING NO.				
SCAFFOLDING WORKS		PMS-WA-ARCO-025-SIDE-VIEW				
DRAWING TITLE		SCALE	DATE	REV	COMPANY	
PMS-WA-ARCO-025 - SIDE VIEW		NTS	07-JUL-2026	0		
CLIENT	LOCATION	DRAWN	REVIEWED	APPROVED	SHEET	SIZE
		Nnamani, Ifeanyi	Ezewoko, Joseph	Chima, Tasieobi	2 OF 4	A3



PLAN VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-025-PLAN-VIEW			 COMPANY	
DRAWING TITLE PMS-WA-ARCO-025 - PLAN VIEW		SCALE NTS	DATE 07-JUL-2026	REV 0		
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi	SHEET 3 OF 4	SIZE A3



3D VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-025-3D-VIEW			 
DRAWING TITLE PMS-WA-ARCO-025 - 3D VIEW		SCALE NTS	DATE 07-JUL-2026	REV 0	
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi	COMPANY
					SHEET 4 OF 4
					SIZE A3